IMPROVING MID-BLOCK CONNECTIONS

SOUTH GRANVILLE SKYTRAIN STATION

SD381 x City Studio Vancouver

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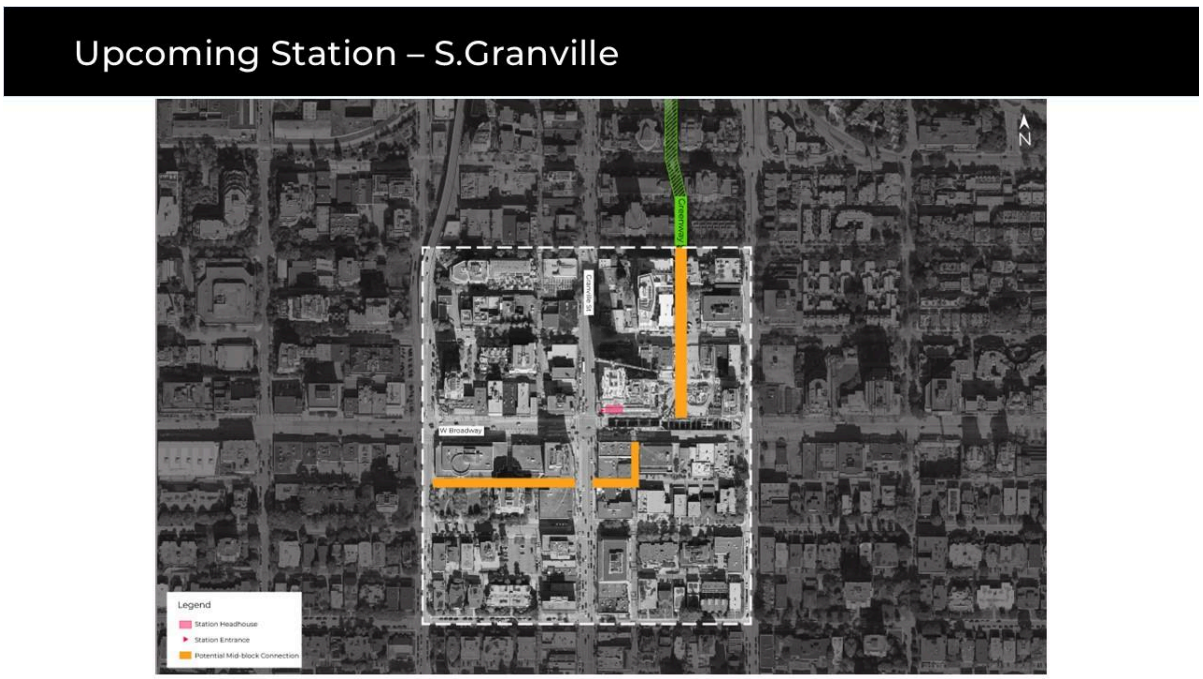
Introduction

Mid-block connections (MBCs) are approximately 6-metre-wide (*Figure 1*) pedestrian pathways between buildings in urban settings leading to key destinations. This report will explore how to redevelop MBCs in the South Granville Skytrain Station area, located at the intersection of Granville Street and West Broadway (*Figures 2, 3, 4*). Design suggestions promote mental wellness and social well-being, focusing on the pre-teen and young teen (ages 12-14) user group. Within that theme, the chapters in this report cover sub-themes of strengthening urban security perceptions, addition of youth-oriented infrastructure, improvement of the physical environment, implementation of natural elements, and promotion of environmental equity. Chapters are ordered beginning with more general redevelopment suggestions in the MBC area, progressing to sections addressing more specific changes within MBCs. Ultimately, this report seeks to lower barriers to pre-teen and young teens' use of public transportation by improving MBCs with their needs in mind.

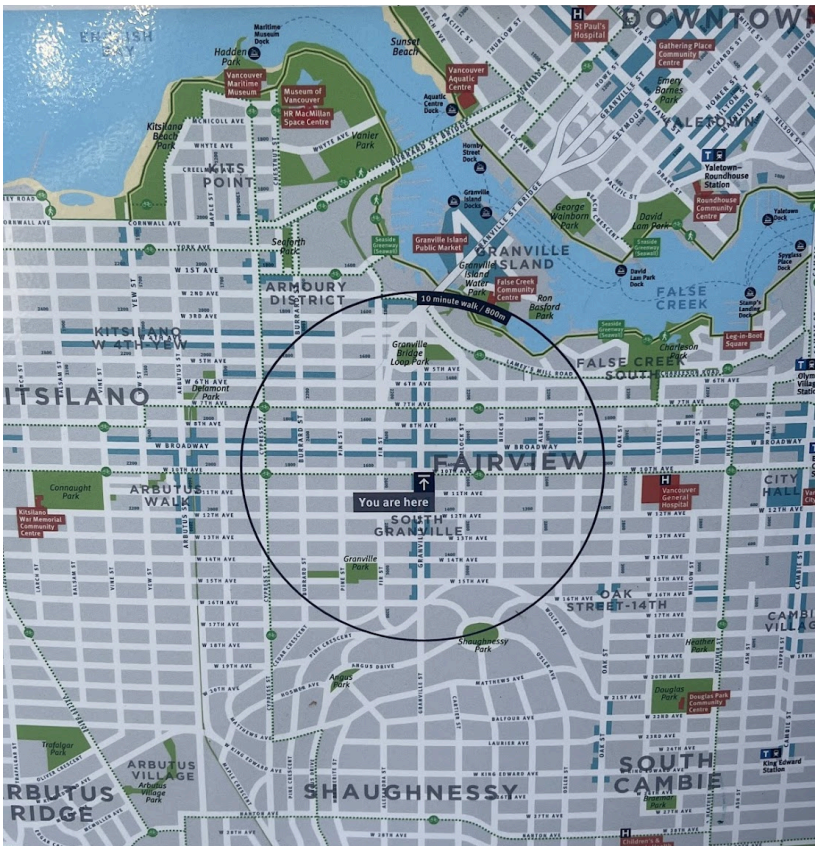
Source: Saenz, 2024

Figure 1. *Typical Vancouver alley (site of future MBC); approximately 6 metres in width.*



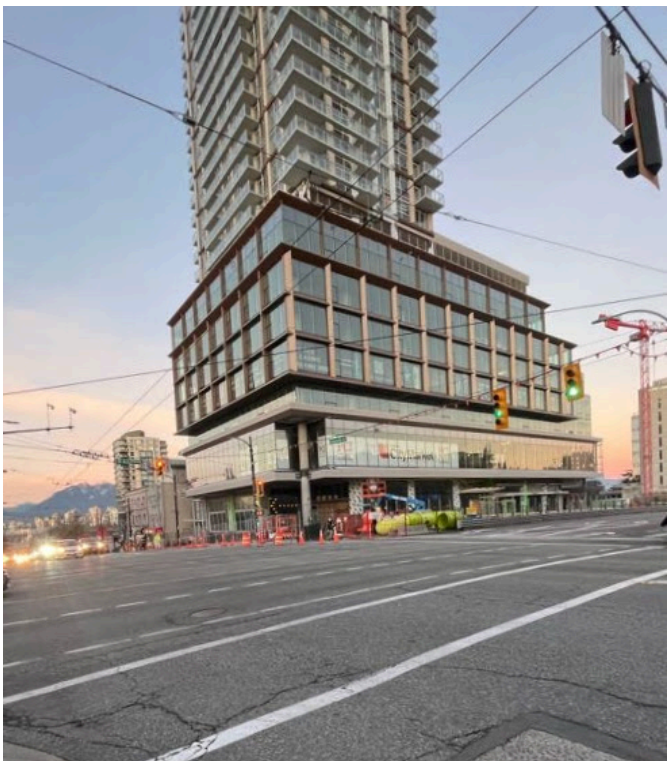
Source: City Studio Vancouver, 2024

Figure 2. *Map of station area and mid-block connections.*



Source: Saenz, 2024

Figure 3. *Map showing further context of station area, relative to other parts of Vancouver.*



Source: Ivkov, 2024

Figure 4. *Intersection of Granville St. and West Broadway, and future site of the station (centre building).*

This report outlines a comprehensive strategy to transform the South Granville Station area into a secure, inclusive, and supportive urban hub for pre-teens, teens, and their families. It begins by addressing safety concerns through energy-efficient lighting, advanced monitoring technologies, and community workshops to foster trust and create a safer transit experience. Building on this, the report advocates for youth-oriented infrastructure, including walkable communities, enhanced library accessibility, and recreational spaces to promote social, mental, and physical well-being.

Furthermore, public art and culturally inclusive designs are proposed to enrich urban aesthetics, improve accessibility, and support mental health. Natural elements, such as lavender-lined paths, rooftop gardens, and green climbing walls, are integrated to encourage relaxation, physical activity, and environmental awareness. Finally, the report highlights environmental equity, addressing pollution and expanding green spaces to improve youth well-being. Together, these measures reimagine South Granville Station as a nurturing space for community and growth.

STRENGTHENING URBAN SECURITY PERCEPTIONS

Maria Fatima Saenz

Strengthening Urban Security Perceptions in Mid-Block Connections Around South Granville Station

Executive Summary

This proposal seeks to fortify urban security at the South Granville Station area by addressing safety concerns for pre-teens, teens, and their families. Despite lower crime rates, public perceptions of safety—particularly among youth and parents—remain low. Key recommendations include installing energy-efficient lighting to improve visibility, integrating advanced technology solutions, and implementing community-driven workshops to educate and engage stakeholders. Together, these measures aim to improve trust, enhance perceptions of security, and create a more comfortable transit environment and experience overall.

Introduction

Security is a fundamental aspect of urban planning, recognized as a basic human need. As perceptions of new and significant threats in urban areas grow, ensuring safety has become a critical priority on city planning agendas. For pre-teens (9–12 years) and teens (13–19 years), safety concerns pose a significant barrier, deeply impacting their mobility and independence. This proposal addresses these challenges by integrating insights from safety statistics, parental concerns, and research on urban security and design thinking. These elements intend to form the foundation for developing targeted measures to strengthen security for the target age-group in the Granville SkyTrain Station area.

Section I: Setting Presentation

Site Visit: South Granville Area Observations

A site visit to the South Granville area revealed that the neighborhood serves as a hub for commerce, culture, and community. Its upscale boutiques, art galleries, and dining establishments were observed to attract both residents and visitors, cementing its importance along the Broadway Corridor.

Demographically, the area is characterized by a predominantly high-income population that includes professionals, families, and retirees, alongside students and young creatives. This affluent demographic presents a key consideration for the SkyTrain project, as their elevated expectations for safety, efficiency, and design are likely to shape perceptions of the station's integration into the neighborhood.

In this framework, the anticipated increase in connectivity and visitor flow from other parts of Vancouver may raise concerns among residents, particularly regarding safety. Parents of pre-teens and teens, in particular, may question how the influx of transit users could affect their community's security for their children. To address these apprehensions, it is essential to examine existing safety statistics and perceptions, which will inform strategies for ensuring the project aligns with residents' expectations while maintaining a sense of security.

Analyzing Safety Statistics for Urban Design

Statistics highlight a persistent gap between crime rates and perceived safety, particularly in urban areas like Vancouver. Despite the city reporting violent crime rates that are 15% lower than the national average, only 31% of its residents express high satisfaction with personal safety—a figure significantly below the national average (38%) (Statistics Canada, 2020). This disconnect highlights a paradox in urban security, where objective crime data fails to align with public perceptions.

Public transit systems further amplify these concerns. Ipsos Canada (2023) notes that safety worries have triggered behavioral changes among Canadians, particularly younger demographics. Statistics Canada (2020) also reveals that younger populations have reported the most significant shifts in their sense of safety since the pandemic, reflecting broader anxieties about urban spaces. While urban areas experienced a 13% decline in overall crime between 2011 and 2021, compared to only a 3% reduction in rural areas, urban spaces are still perceived as more dangerous, shaping public behavior and decision-making (Perreault, 2021: p. 3).

These insights emphasize the need for urban designs that bridge the gap between perception and reality. Aligning youth safety perceptions with actual crime statistics requires targeted environmental improvements, such as better lighting and visibility, and community-driven engagement strategies to assure positive security perceptions. The greater the sense of safety among transit users, the higher their satisfaction, which will drive increased usage of transit services, creating mutual benefits for both customers and service providers.

Section II: Foundations for Security Measures

The Urban Security Framework

In *Urban Security as part of the Smart Cities Strategy,* Wladyslay et. al (2010) states that "the security feature of urban space is the basis of the good functioning of cities and is a significant factor in creating smart city strategies" (p.2). In line with this, the challenges that

arise from progressive urbanization are shown to have increased the social demand for more efficient and sustainable urban services (p.3).

Therefore, the authors draw that ideally functioning cities are conditioned by the degree of ensuring security needs and cannot function without it (p. 4). This thesis proposes that achieving security in urban spaces depends on preventive measures grounded in architectural strategies, particularly within the spheres of settlement systems, public spaces, and information technology (p.5). However, achieving this requires the integration of design thinking methods to address the group's perceptions of safety in the selected South Granville Station mid-block connection.

Applying Design Thinking

Applying Dekker's (2020) insights on empathic design can significantly enrich the present proposal. In *Design Thinking is a way of thinking,* the author states that in order to "create coherence, there is a need for working integrally through cooperation and connection with others" (p. 27). Similar to Dekker (2020), Masys (2003, p. 25–26) argues in *Safety and Security/Science and Technology* that this method has proven effective in leveraging people-centered approaches across various projects.

Moreover, Masys emphasizes the importance of applying empathy to fully grasp that effective security planning necessitates immersing oneself in the community's lived experiences. From these perspectives, focusing on the insights of pre-teens, teens, and parents is central to taking a user-centered approach that fulfills creating an inclusive space that prominently strengthens security.

Section III: Integrating Theory into User-Centered Approaches

Urban planning emphasizes two key dimensions of security: (1) freedom from threats and (2) the assurance of survival and growth (Wladyslaw et al., 2010, p. 1). These dimensions are critical for youth, as their perception of safety profoundly influences their mobility and overall well-being.

An important variable to consider in this analysis is parent's perceptions on security for their children. Wilson's et. al. (2018) study on *Understanding child and parent perceptions of barriers influencing children's active school travel* suggests that "parent perceptions of barriers are found to have a greater influence on children's travel behaviour than children's

perceptions" (pp.1). For young transit users and their parents, the perception of safety must encompass more than just the technical absence of crime—it must also be visually apparent.

Lighting Improvements to Enhance Visual Safety Perceptions

One of the most effective measures to enhance safety is improving lighting throughout the station. Installing high-quality, energy-efficient LED lighting can eliminate dark or shadowed areas **(See Figure 2)**, making the surroundings feel more inviting while reducing opportunities for crime. Additionally, designing open spaces with clear sightlines enhances visibility and minimizes hidden areas where potential threats might arise. For youth specifically, incorporating designated gathering zones can significantly enhance their experience and build parental trust in their ability to navigate transit independently.

Collectively, these measures can positively influence parents' perceptions of security. This, in turn, will also improve the age group's perception of safety and comfort. By addressing these concerns, the South Granville Station area, and the selected mid-block connection near it can see increased usage while becoming a more inclusive, secure space for all. Further analysis on lighting approaches will be discussed in the *Improvement of Physical Environment* chapter.

Source: Saenz

Figure 1: Alley Nearby the Granville Station Area Project

Technology Incorporation

Integrating technology is another crucial step in enhancing urban security. The first aspect that has to be considered is the improvement of quality surveillance in the area. An example of this are real-time monitoring systems, such as AI-enabled networks. These can deter criminal activity and provide rapid responses to potential incidents. These cameras should focus on high-traffic zones, entrances, and exits to ensure comprehensive coverage **(See Figure 2).**

On the other hand, a mobile safety app tailored specifically for young users could prove to be an invaluable tool. Such an app might include features like live transit tracking, emergency alerts, and a direct communication channel with station security personnel. According to Statistics Canada's (2022) *Canadian Internet Use Survey*, as digital technologies continue to advance, Canadians are increasingly integrating technology into their daily lives. Internet usage among Canadians aged 15 and older has reached 95%, highlighting the potential for widespread adoption of such tools across young and older generations.

Incorporating a safety app would streamline daily routines for both parents and young users by providing real-time updates and alerts about potential threats or changes in their environment. Complementing this idea, interactive digital kiosks installed at the station could offer supplementary resources such as safety tips, app instructions, and transit information. However, this measure needs to be supported by community engagement dynamics to come out effectively.

Source: Saenz – Figure 2: South granville 12th Ave.

Community Ownership Dynamics

The finishing touch to this security enhancement proposal is to strengthen the culture of safety features through community engagement dynamics. To start, school workshops tailored for teens can educate them about navigating transit systems, recognizing risks, and responding to emergencies. The core of these is to provide guidance and support to increase community ownership.

However, pre-teens and teens may not fully recognize the importance of this space. Blakemore's (2018, p. 116) research on *Avoiding Social Risks in Adolescents* highlights that they often resist structured programs like workshops, perceiving them as imposed and conflicting with their desire for independence. This tendency could lead to resistance toward the proposed workshops. In this context, Ryan and Deci's (2000, p. 76) study on *Self-Determination Theory* suggests that teens are more likely to engage in workshops when the activities foster their sense of autonomy, competence, and relatedness.

To minimize resistance, strategies should focus on co-creating workshop topics with participants, integrating interactive and enjoyable elements, and offering practical incentives such as transit passes or certificates with real-world value. The involvement of law enforcement and security agencies in these workshops can significantly enhance the security perceptions of both youth and parents. By educating participants about their role in ensuring public safety and introducing new measures, such as increased presence during peak youth transit hours, these agencies can make a meaningful contribution to the initiative's success. Their active participation in these events can reinforce a sense of safety and trust among attendees.

After establishing initial engagement, creating accessible feedback channels—such as surveys or suggestion boxes—can provide teens and their families with opportunities to voice concerns and share ideas. This inclusive approach fosters a sense of community ownership over the initiative, encouraging continued participation and strengthening motivation to contribute to its development.

Conclusion

In conclusion, this proposal emphasized the critical importance of addressing urban security perceptions to develop effective measures, particularly for pre-teens and teens as the target demographic. By evaluating key statistics and integrating comprehensive research on urban security and design thinking, this chapter establishes a rich framework for aligning architectural strategies with the specific needs of this age group. These measures aim to progressively address and reshape their perceptions of safety, recognizing their significance as valuable transit system users.

The subsequent chapters will delve into crucial aspects not covered here, including specific improvements to mid-block connections through physical environment improvements, the integration of natural elements, and strategies for achieving environmental and spatial equity through public space activation. Together, these components strengthen safety perceptions by fostering visually and functionally improved urban mid-block connections.

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ENHANCING EQUITY THROUGH PUBLIC SPACE ACTIVATION

Kiranpreet Gill

Enhancing Equity through Public Space Activation in Mid-Block Connections Around South Granville Station

Executive Summary

Midblock connectivity refers to the ease of movement and access between streets or locations that are not at major intersections, typically in the middle of a block. It involves creating pathways, sidewalks, or other routes that allow pedestrians, cyclists, or vehicles to move efficiently and safely between different areas, without needing to rely on busy intersections (Vancouver Public Space Network, 2017). This chapter advocates for the development of youth-oriented infrastructure around the South Granville Skytrain Station in Vancouver to improve the mental, social, and physical well-being of pre-teens and teens. Key features include creating walkable communities that encourage active transportation, enhancing the accessibility of the Vancouver Public Library, and adding a park and recreational facility. These initiatives aim to provide safe and engaging spaces that encourage social interaction and physical activity.

Introduction

Young people engage with and experience their surroundings differently from adults, frequently relying more on their local environment (Nissen, Prendergast, Aoyagi, et al., 2020). By promoting connected, multifunctional, and sustainable public spaces for the South Granville Station in Vancouver, youth infrastructure can generate considerable benefits for all citizens, especially in child-friendly and sustainable towns, as recommended by Nissen et al., (2020). This can be accomplished in various ways, like constructing walkable communities that support public transit, designing an accessible public library, developing an open green space, and building a recreational facility that supports multifarious sports and activities with indoor and outdoor gymnasiums (Thomas & Bertolini, 2020).

Walkable Communities

Baobeid, Koç, and Al-Ghamdi (2021) define walkable communities as the quality of built environments that enable the mobility of pedestrians. This includes the necessity for sidewalks that are accessible, connected, safe, comfortable, shaded, and well-maintained for the full spectrum of pedestrians (Baobeid, Koç, & Al-Ghamdi, 2021). Walking is a low-intensity physical activity associated with healthier populations since it contributes to

lower obesity rates, diabetes, and respiratory diseases, especially among adults (Baobeid et al., 2021). However, pre-teens and teenagers contribute significantly to the population that avoids walking and cycling and, instead, travels by vehicle whenever possible (Baobeid et al., 2021). This can be the result of many factors, including severe weather conditions, time constraints, or, simply, disinterest. To resolve this issue, safe and aesthetically pleasing walkable communities should be incorporated into the busy city to generate and address heavier foot traffic along the midblock connections. This can be accomplished by planting Evergreen shrubs along the sidewalk of Granville Street to provide shade, improve air quality, and further separate pedestrians from the busy roads while ensuring their visibility to local traffic (O'Donnell, Knab, & Athey, 2007). Incorporating greenery into accessible walking areas further encourages individuals to utilize safe and engaging sidewalks that efficiently connect to their destination.

Additional Park and Recreational Facility

Urban parks are defined as open space areas that are mostly dominated by vegetation and water and are generally reserved for public use (Konijnendijk, Annerstedt, Nielsen, & Maruthaveeran, 2013). They have a positive impact on human health and well-being through both direct and indirect effects such as recreation and leisure activities, as well as strengthening social ties and reducing stress (Konijnendijk et al., 2013). This is seen through the reduction of anxiety and mental disorders in young adolescents by providing opportunities to increase their physical activity and social interaction. An urban park along West Broadway Street will provide a site for pre-teens and teens to engage with their community, foster a sense of belonging and positive mental development, and allow them to easily access nature to further combat urban stressors (Konijnendijk et al., 2013). The park can feature sheltered areas that encourage socialization and relaxation, like benches and tables, as well as areas for physical activity, such as a basketball court and soccer field. Regarding midblock connections, the park would enhance pedestrian and cyclist mobility by supplying a secure and accessible pathway to cut through the area and reach the SkyTrain Station more efficiently. This design would not only improve connectivity but also create a vibrant, multifunctional space for the community.

Figure 1: Park. (n.d.). *Photograph of a park*. Retrieved November 15, 2024, from https://en.wikipedia.org/wiki/Park

From 1988 to 1989, a large sample of 9th-11th grade public high school students in the United States were questioned twice about their sports involvement and mental health, with data collected one year apart (Feng, 2022). The students who participated in athletics showed decreased levels of depression, compared to the students who rarely or never participated in sports. These results indicate that partaking in sports promotes emotional balance for teenagers, and the study supports the benefits of sports participation for both normally developing and disabled adolescents. This study highlights the necessity for a recreational facility near South Granville Station that is accessible to all children, adolescents, and adults in the area who are seeking physical activity to improve their mental and physical health, increase their social interaction, develop new skills, and provide a space for them to relax and have fun. The presence of a recreational facility inherently promotes a more active and pedestrian-friendly environment by serving as a destination that motivates people to choose walking or cycling over driving. It would appeal to pre-teens and teens seeking fitness opportunities, social engagement, or leisure activities—all of which align naturally with active transportation methods (Feng, 2022).

By placing this facility within a walkable distance from the Skytrain Station, individuals can reduce their reliance on cars, cut down on traffic congestion, and lower greenhouse gas emissions. Many researchers have studied the relationship between facility availability and physical activity behavior in kids, utilizing both perceived and objective measures of facility

availability. The majority of this research found that the higher availability of facilities in an urban area correlates with increased physical activity for youth and adolescents (Ries, Yan, & Voorhees, 2011). This is because if there is an increased opportunity for individuals to participate in physical activity, they are more likely to take advantage of that opportunity, rather than dismiss it, especially if the former option is more financially and regionally accessible (Ries et al., 2011). This demonstration of a favorable relationship between availability and physical activity further indicates the need for a facility in the area, especially one that can be accessed before and after a train ride or any trip to a destination.

Recreational facilities have typically been categorized into several classifications, such as basketball courts, bike trails, and walking paths, which are associated with their success in attracting individuals to the facility. However, the mere presence and number of recreational facilities in a neighborhood may not be sufficient to encourage physical activity, as other characteristics like admission prices and adequate lighting can also influence their utilization, as well as the varying effects of indoor and outdoor facilities. Outdoor installations are restricted by weather and safety issues, which make indoor activities seem desirable in certain situations. In contrast, although indoor activities are immune to the influence of the external environment, they may be less favorable to external aesthetic features, like natural landscapes (Lee, Lee, & Macfarlane, 2014).

With this in mind, the recreational facility should feature indoor and outdoor sports courts for activities such as basketball, volleyball, and tennis, and a gymnasium with fitness classes, durable equipment, and sports gear. A game room with pool and air hockey tables, a dance studio, and a lounge area would further facilitate social interactions among the guests. Additionally, ensuring the building is equipped with high-speed wi-fi, efficient air conditioning, healthy snacks, and trained staff who are qualified to assist in situations when first-aid is required will further enhance the quality of the center. Finally, the facility will provide numerous job opportunities for individuals of all ages and experience. The teenagers can work in the cafe to earn valuable customer service and cash handling skills and manage the lounge areas when needed. Alternatively, experienced individuals can supervise the outdoor and indoor gymnasiums, and provide first-aid knowledge when needed for safety and security.

This recreational facility would significantly enhance the midblock connection by providing a direct and accessible pathway that encourages pedestrians and cyclists to use the space. By

strategically integrating sports courts, a gymnasium, and social areas, it would not only serve as a destination but also act as a connector, improving flow through the midblock area (Lee, Lee, & Macfarlane, 2014). Well-designed entrances and pathways would create clear and safe routes for people to traverse, which would also link the neighborhood and Skytrain Station more efficiently. The addition of amenities like bike racks, shaded seating, and proper lighting would further enhance the safety and convenience of the midblock connection, ensuring that people feel comfortable walking or cycling through the area.

Figure 2: City of Vancouver. (2023). *Creekside Community Recreation Centre* [Image]. Retrieved from https://vancouver.ca

Enhancing Access to the Library

A public library's purpose is to promote enlightenment, education, and other cultural activities through the dissemination of information, and by making books and other suitable materials available to the public, free of charge (Aabø, 2005). They provide easy access to resources like books and the internet, and study areas that encourage learning. Making the Vancouver Public Library more accessible can transform it into a central destination that actively encourages greater use of the surrounding midblock pathways. This can be achieved through thoughtful design improvements, such as adding a ramp and widening the sidewalks from the Skytrain Station to the library. Installing a ramp alongside the stairs of the entrance to the library ensures that individuals with mobility impairments can easily access it, while wider sidewalks can accommodate both pedestrians and cyclists, promoting more inclusive

transportation options. These upgrades not only enhance the library's accessibility but also encourage greater foot traffic through the midblock space, as people are more likely to visit the library if they can do so easily and safely, especially young teens.

Conclusion

Developing youth-oriented infrastructure around the South Granville Skytrain Station in Vancouver can contribute to the mental and social well-being of pre-teens and teens in the area, by creating opportunities for physical activity and social well-being. Designing accessible and connected sidewalks that are secure, well-maintained, and aesthetically appealing, will encourage the target demographic to walk or cycle to the train station, as opposed to relying on a private vehicle (Baobeid et al., 2021). In addition to walkability, the presence of a park and recreational facility can considerably enhance their mental health through relaxation, socialization, and physical activity. Lastly, equipping the public library with safer and more engaging walking and accessibility features will make it more approachable for individuals with mobility challenges. Ultimately, the combination of a walkable, sustainable space, recreational facility, and accessible library near the transit station will support this demographic's mental, physical, and social health by offering them a space to engage in activities that will promote their healthy development.

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IMPROVEMENT OF PHYSICAL ENVIRONMENT

Smaye De Villa

Improvement of Physical Environment in Mid-Block Connections Around South Granville Station

Executive Summary

The subsequent section will focus on aspects of physical environment design that can be implemented in a mid-block connection located in the South Granville Skytrain Station area. The key topics include the implementation of public art, the role of aesthetics through architectural and street design enhancements and strategies to increase inclusivity. Public art, such as murals and interactive installations, can serve as a tool to reduce stress, increase adolescent interaction within the space, and provide cultural representation for diverse groups. Architectural elements, such as facades with more complex designs and vibrant colors, can positively impact emotional well-being. Meanwhile, inclusivity strategies include multilingual signage and accessible infrastructure for adolescents with disabilities. The report includes evidence based on academic studies and examples of how these ideas could be implemented in this specific mid-block connection area.

Introduction

Mental health and social well-being are vital aspects to achieve a sustainable design. This can be accomplished through various strategies. The following study focuses on approaches that can be specifically employed for the demographic group of pre-teens and teens in the area surrounding the South Granville Skytrain Station (Figure 1), aiming to also promote the use of active and public transportation, as this group may face several barriers to use it. According to a study, one of the barriers of active travel for adolescents is the aesthetics of the physical environment, as it significantly impacts their sensory experience. For instance, environments that are more aesthetically pleasing can enhance the perception of active travel, making it feel more enjoyable (Buttazoni, Nelson & Gilliland, 2023). Beyond aesthetics, this paper will explore additional factors that have the potential to influence adolescents' experiences in the mid-block connection, and consequently, their emotional well-being derived from these interactions.

Source: City Studio Vancouver

Figure 1. *Suggested area for mid-block connection located around Granville Street and W Broadway intersection.*

Public Art

One way to address adolescents' emotional well-being is through art. A study demonstrated that art incorporated into urban design has the potential to improve people's emotional states, reducing anxiety and stress (Mikuni et al., 2024). One of the essential findings of this study is that even incorporating artistic elements such as laminated art prints on streets, have the same effect as with natural elements, which have also been proven to reduce stress and anxiety levels (Mikuni et al., 2024). This discovery is particularly useful in spaces where incorporating or maintaining natural elements may be difficult. For example, in the specific hallway of the block connection shown in **Figure 2**., the space is somewhat limited, so fewer natural elements can be included. Additionally, in winter, many of these natural elements lose their leaves and vibrant colors, so methods like art on the walls could continue to support the positive effect intended for adolescents' emotional well-being. For these reasons, public art could complement the biophilic approach that will be explored in the following section.

Art can also help adolescents increase their sense of belonging to the place. One idea would be to incorporate murals on the walls of this hallway with different themes. For example, in the Lakeview community in Chicago (see **Figure 3.**), public art has been implemented as an alternative to raise awareness of various issues, including mental health (Lakeview Roscoe Village Chamber of Commerce, 2024). Additionally, this city has also implemented programs to encourage adolescent participation in public art projects, in collaboration with professional artists. A program like this could be included in this area of Vancouver to allow adolescents to participate in the creation of art, helping them feel they have a voice in their community. These programs could be promoted in schools to encourage engagement and ensure that the art reflects the perspectives and identities of the youth involved. The aim is to promote the process of place-making, through which residents can feel a connection with their community environment and have the power to express themselves through art (Chicago Public Art, n.d.). Additionally, this involvement of adolescents in creating public art can also serve as a form of therapy. According to a study exploring the relationship between the role of art and adolescents' mental well-being, it was found that art can help them learn to express their feelings, manage stress, and, in some cases, improve self-confidence (Mittal, et al., 2022).

Source: Google Earth

Figure 2. *Proposed alley for art implementation*

Source: Lakeview Roscoe Village Chamber of Commerce. (2024). Public Art Program. https://www.lakeviewroscoevillage.org/public-art-program

Figure 3. *Public Art Program*

As stated by Dr. Lace Padilla (2021) choice architecture refers to the system designed to influence the outcome of people's decisions by altering how the options leading to a particular action are presented. Choice architects, therefore, must create stimuli to encourage individuals to take a specific action (Dr. Lace Padilla, 2021). In this case, the stimuli in the environment should be designed to promote interaction with art, which is the desired action. According to some principles of choice architecture, certain defaults should be considered to encourage adolescents to engage more with public art. For example, due to Vancouver's climate conditions, known for constant rain, it would be ideal to have a default setting where public art installations are accompanied by nearby shelters so that rain doesn't prevent people from interacting with the art. For example, in the corridor where wall murals were proposed, a transparent roof could be included to protect against the rain. This would ensure that the space remains accessible and inviting, allowing people to interact with the art comfortably regardless of the weather. There should also be incentives to ensure that rain doesn't affect this interaction. For instance, there is interactive art designed to function in the rain, like "Courtyard of Water" (Figure 4), an art installation on a building wall in Berlin, Germany, which produces musical sounds when it rains and also serves as a water collection system (Davis, 2017).

Source: Davis, T. (2017). Sculptures that entertain when it rains - Places for Everyone. Places for Everyone. https://placesforeveryone.com/index.php/2017/02/01/sculptures-that-celebrate-rain/

Figure 4. *Interactive art under rain conditions*

Another principle of this choice architecture is to anticipate errors. In this case, it is expected that not all art will appeal to everyone, which may hinder interaction and its potential benefits. Factors such as individual preferences or cultural backgrounds can influence this, as individuals from different cultures might not feel connected to certain art. This is why there should be a diversity of art, including pieces that are more representative of adolescents belonging to First Nations communities. For example, a valuable feature of this area is the totems located in the park adjacent to the Block Connection (shown in Figure 5.)

Source: Google Earth

Figure 5. *Indigenous Art Implementation*

Aesthetics

Another important aspect of urban design that influences the emotional state of adolescents is the complexity of the urban environment in which they are located. According to a study, complexity is defined as the visual richness provided by urban elements like architecture. A positive relationship was found between the diversity of architecture, its colors, and styles with the emotional state of adolescents, as it increases their engagement with their surroundings (Buttazzoni & Minaker, 2023). Another study that also explored the relationship between architecture and people's emotional state found that some architectural aspects can generate negative emotions like sadness and displeasure, such as poorly maintained buildings or very tall new structures, which can block natural light (Zumelzu et al., 2024). The figures below demonstrate how some parts of the buildings in the mid-block connection are poorly maintained in certain areas **(Figure 6)**, and how some buildings are so tall that they block natural light in certain corridors **(Figure 7)**. In areas where tall buildings block sunlight, there should be some form of artificial lighting as suggested in the strengthening urban security section. These should be powered by solar panels to ensure that, when it gets darker, adolescents do not feel unsafe.

Conversely, this implementation can have negative effects on wildlife, as it may disrupt their circadian or biological clocks, which refer to natural behavior patterns during the day, such as sleep (Aulsebrook et al., 2021). Research shows that artificial light can alter the sleep patterns of species like birds, causing them to fall asleep later. Therefore, efforts should focus on minimizing the impact of light. For instance, studies suggest that long-wavelength light has less impact compared to short-wavelength light (Russart & Nelson, 2018). LED-based white lights, for example, tend to fall within the blue spectrum, which is shorter in wavelength. In contrast, long-wavelength lights, such as those in the red spectrum, have a lesser effect on the circadian clock. Using motion sensors to limit light exposure to shorter periods can also reduce its impact on species, thereby mitigating these effects (Aulsebrook et al., 2021). Such considerations could help minimize potential negative consequences for biodiversity while still achieving the primary goal of enhancing the perception of safety. Preserving biodiversity is essential as it supports the attraction of potential pollinators to the mid-block connection area, an idea further explored in the next section on the implementation of natural elements.

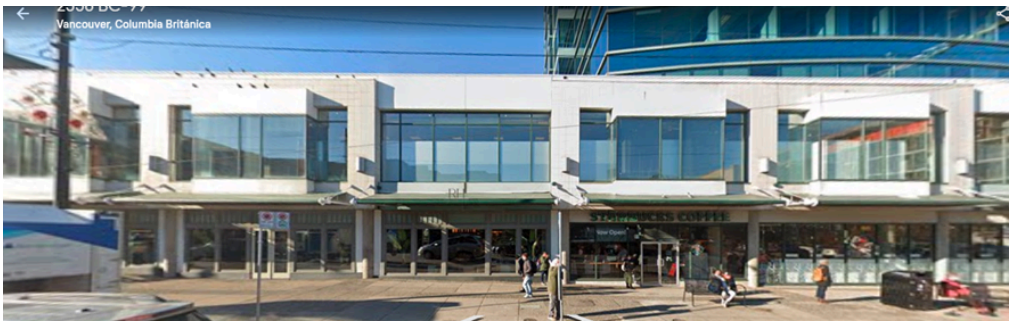
Source: Google Earth

Figure 6. *Building with poor maintaining conditions*

Source: Google Earth

Figure 7. *Tall building blocking sunlight in alley*

On the other hand, features like a variety of vibrant colors or ornaments on doors and windows on the facades of buildings can foster feelings of community identity and enjoyment (Zumelzu et al., 2024). Several buildings in this area (**Figure 8)**, for example, lack a great diversity of colors or architecture styles, increasing a sense of monotony. According to Colin Ellard, a psychology researcher from the University of Waterloo, building facades that are simple and monotonous, can affect people negatively, preventing them to engage more in those areas.



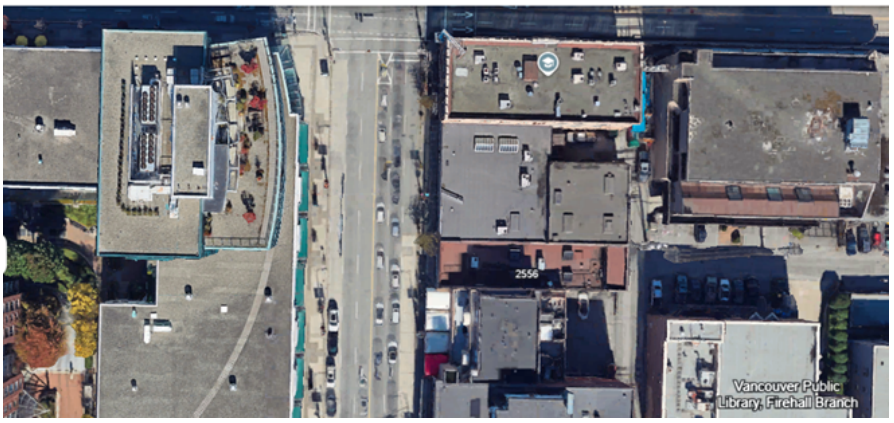
Source: Google Earth

Figure 8. *Building with monotonous facade*

Another aspect of urban design aesthetics that can promote people's well-being is the colors and patterns of sidewalks. A study states that these elements can influence people's engagement with the space, as they enhance walkability and interest in the area (Gu et al., 2024). The study proposes the implementation of ground murals, as they contain key elements for stimulating positive emotions. For example, it was found that cool colors like blue and green can be restorative while warmer colors can decrease relaxation, while achromatic colors do not have any positive or negative effects. Leon Rainbow is a community project in which several floor murals were implemented in Washington (Figure), primarily to serve as calming features for traffic, thereby increasing pedestrian safety and supporting mental well-being too (Leon Rainbow, n.d.). Consequently, these future floor murals can achieve multiple objectives. The **figure 10**. shows one of the possible locations where they could be implemented, specifically at the crossing from one street to another. However, the location of these must be well chosen to avoid saturation of the elements.

Source: Leon Rainbow. (n.d.). https://www.leonrainbow.com/ground-murals/

Figure 9. *Example of floor murals*



Source: Google Earth

Figure 10. *Proposed section of mid block connection for floor murals*

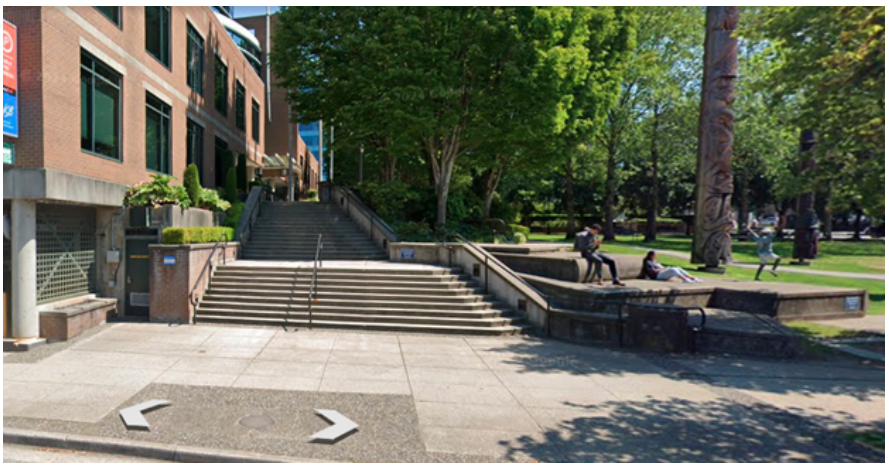
Inclusivity

Despite the similarities in the barriers and needs faced by adolescent groups, it is also necessary to account for the differences within this group to design cities that are inclusive for everyone. One method previously mentioned was using art to represent adolescents from specific communities, fostering inclusivity. Another way to promote inclusivity and representation is through the use of language, such as multilingual street signs. In this case, incorporating the languages of First Nations communities would be appropriate, as Vancouver is located on the traditional territories of the Musqueam, Squamish, and Tsleil-Waututh peoples. Due to historical contexts, many individuals have lost their connection to these lands and often feel excluded from these spaces. For instance, an example could be illustrated in a bilingual sign project to honour the Ucluelet First Nation, making members of this community feel more respected and included in the area, shown in **Figure 11**. (Wilson, 2022).

Source: Wilson, C. (2022, July 29). Ucluelet gets bilingual street signs: Nuu-chah-nulth plus English. Times Colonist. https://www.timescolonist.com/local-news/ucluelet-gets-bilingual-street-signs-nuu-chah-nulth-plus-english-5638171

Figure 11. *Cultural representation examples through street signs*

Another way is to ensure that the physical environment is accessible to all, including those with disabilities. For example, according to the City of Vancouver, future corridors in the mid-block connection must have a minimum of 1500 mm to ensure that the space is sufficient for people in wheelchairs or with visual impairments that require a guide dog (City of Vancouver Engineering Services, 2019). Another aspect is for example to have places to sit, as some people with disabilities require constant rest every certain distance depending on their needs. This requirement might conflict with the need for more space, so it is important to strategically decide where to place these benches. One possible location is the park next to the connection, and make sure there are enough benches in close proximity to the stairs. Sufficient ramps should also be provided for those in wheelchairs as this section only includes stairs and sitting areas are not accessible for teens with wheelchairs (Figure 12).



Source: Google Earth

Figure 12. *Proposed area of mid block connection for infrastructure improvement*

Conclusion

Improving the physical environment has the potential to improve the mental health and well being of pre teens and teens by addressing key aspects such as the implementation of public art, aesthetic improvements in architectural and street design, and the promotion of inclusivity.All these suggestions are supported by evidence from other studies and cases, serving as a foundation for their implementation in this area of Vancouver to enhance the mental health and well-being of adolescents and pre-teens and contributing to the goals of building more sustainable communities that address the needs of everyone.

To ensure the success of these recommendations it is also important to increase youth participation in the decision-making process for the construction of this mid-block connection to ensure that all their needs are addressed and to foster a sense of connection with the space. Additionally, regular monitoring of adolescents' mental health and well-being is suggested to evaluate the success of specific elements and adapt them as needed to better achieve the intended goals.

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IMPLEMENTATION OF NATURAL ELEMENTS

Natasha Ivkov

Implementation of Natural Elements into Mid-Block Connections Around South Granville Station

Executive Summary

A key aspect of enhancing mid-block connections (MBCs) around the South Granville Station area, in terms of mental wellness and social well-being, is implementing elements of nature into them. This section proposes implementation of natural elements that appeal to suburban pre-teens and young teens ("tweens") who use these MBCs. However, these features serve everyone who passes through the corridors, and ultimately create a more sustainable local community. Details like tween interactions with the designs and considerations for seasonal usage are discussed. Evidence behind nature's impacts on mental wellness and social well-being is also incorporated. Depending on the physical feasibility of these ideas, some are ideas for implementation in the alleyways themselves, and some are ideas for the buildings surrounding them. Either way, key principles of these designs are empathic design thinking and choice architecture. Designing intentionally while considering tweens' needs ensures that designs do end up being used. While it may be a challenge to introduce greenery into currently concrete alleyways, it must be a focus of redevelopments made in these MBCs.

Introduction

The following section discusses redevelopment of MBCs in the South Granville Station area through implementation of natural elements. The aim of the redevelopments is to increase mental wellness and social well-being of suburban tweens using these corridors. As mentioned in the full report's introduction, MBCs are about 6 metres wide, which is considered in design suggestions. This chapter discusses topics including relaxation, socialization, cognitive function, spirituality, physical health, and environmental education. This section builds on the previous chapter by Gill, which discussed activities and opportunities for socialization, and the subsequent chapter by Lee builds on this one by discussing the importance of equitable access to nature.

Nature and Mental Wellness

Several mental wellness benefits are associated with being in nature. For tweens dealing with the stresses of school and emotional changes, this is especially relevant; time spent in nature must be a part of their everyday lives.

Relaxation

First, nature has a relaxing effect on humans; time spent in nature either actively or passively is thus mentally beneficial (Schertz et al., 2021). As well, a study by Owens and Mckinnon (2009) shows that nature is among the favorite places for youth specifically to spend time. This study showed that a key portion of this time is spent in "restoration," or psychological rest and relaxation. A way to design for relaxation in MBCs is by introducing sage- or lavender-lined pathways, as these plants are widely used for their calming scents (*Figure 1*). Including signage with interesting facts about these plants (like their history and calming properties) would be engaging for tweens, too.

In addition to time spent alone in nature, Owens and Mckinnon (2009) show that young people highly value socialization. To support this need, South Granville's MBCs could be designed to allow tweens to spend time either alone or together in nature, bringing them multifunctional benefits A key feature of the MBCs should be simple, well-maintained, grassy patches (*Figure 2*). In summer, tweens can gather here, and in winter (specifically in the snow), these areas can be designated for building snow structures (*Figure 3*).

Photo Credit: Pinterest

Figure 1. *Lavender-lined pathway.*

Photo Credit: Natalia Stefan, Dreamstime

Figure 2. G*rassy patches; to fit in an MBC (and leave room for other elements), smaller, square patches of grass could be included along one building, as opposed to the way they are organized in this picture.*

Photo Credit: Kat, Flickr

Figure 3. *Snowbanks in alley; as with the grassy patches, to fit in an MBC, there will realistically only be room for a few areas where snow can be collected for tweens to build snow structures. This would be unlike in this picture, where the entire alley is filled with snow.*

Cognitive Function

Similarly to the point on relaxation, time spent in nature can aid cognitive function. This applies particularly after one has been focusing for a long period of time, like a structured school day involving extensive screen time. Having even more screen time after school, while playing video games, for example, is detrimental to tweens' brain function. In contrast, experiencing nature in MBCs before or after school will provide tweens with the opportunity to rest their brains, equipping them with the mental capacity to continue being productive (Berto, 2005; Bratman et al., 2012).

Spirituality

A final mental wellness-related benefit arising from time spent in nature is the spiritual effect that it has. Time spent in nature gives people the sense of being connected to something larger than themselves, which can renew their sense of purpose (Berejnoi et al., 2019). Feeling connected to nature also reduces feelings of individualism and loneliness, increasing

community and overall happiness (Berejnoi et al., 2019). For tweens who are just beginning to discover their identities, having time to reconnect with nature is crucial—even if it is just briefly in an urban corridor leading to a SkyTrain station.

A design idea encouraging overall mental wellness for tweens in these MBCs is building semi-enclosed capsules for tweens to spend time in within the corridors (*Figures 4, 5*). Designed with Vancouver's rainy seasons in mind, these capsules may include frost-proof seating, small garden plots with perennial (long-lasting) plants inside (potentially protected by clear coverings), and signage or QR codes linked to pages explaining the environmental benefits of these plants. The purpose of these capsules is to give tweens places to decompress in nature, without forcing them to be exposed to the elements on cold days. The sensory experience of seeing, listening to, and smelling the rain may have a relaxing and spiritual effect on users. The roofs of these capsules could also be retractable, allowing them to be opened and used as regular seating spaces during dry weather.

Photo Credits: Tensile Fabric, Architonic

Figures 4, 5. *Examples of semi-enclosed capsules; ideally, covers would be retractable, and capsules would fit in the width of the MBCs with room to pass by on at least one side. As well, they would include more of the elements mentioned, like plants and signage.*

Nature and Social Well-being

In addition to the various mental wellness benefits, there are vast benefits to social well-being brought on by time spent in nature.

Physical Health

The first of these benefits connects to the previous section; increased mental wellness increases physical wellness. Feeling overworked and stressed causes physical symptoms, so reductions in stress benefit physical health, too (Bratman et al., 2012; Schertz et al., 2021); the chapters by Gill and Lee expand on this. Another important physical health-related benefit arises from doing exercise in green MBCs. Public transit use encourages active transport (as opposed to personal vehicle use), so suburban tweens using these corridors to get to the SkyTrain will innately be bettering their physical health through exercise. It is of added benefit that they will be in nature, since physical activity done in nature is proven to have an even larger positive impact on mood than other forms of exercise (Schertz et al., 2021; Taylor & Hochuli, 2015, p. 751).

Active transportation and green exercise in this region are highly supported by the northernmost MBC's proximity to a greenway, and the greenspace adjacent to the easternmost MBC. When tweens encounter the MBCs on their routes, they must feel that the MBCs themselves encourage active transport and green exercise, too. Walking, jogging, cycling, or rollerblading are examples of modes of active transport that tweens could choose to move through MBCs. It is thus crucial to have designated paths within the MBCs for transportation, made of a durable, shock-absorbing recycled rubber, to support physical activity and the environment (*Figure 6*). While in the MBCs, tweens could also participate in other forms of low-impact green exercise. The addition of stepping stones in the form of an obstacle course (*Figures 7, 8*) or low-to-ground green climbing walls (*Figure 9*) could make tweens feel like they are truly immersed in nature while exercising, boosting physical and mental health.

Experiencing these physical health benefits will make tweens healthier in the short and long term—this is a crucial aspect of a sustainable city (Tate, 2024). The section by Lee will

expand on the importance of equitable access to physical health benefits in urban environments.

Photo Credit: Rubberway

Figure 6. *Rubberized pathway.*

Photo Credits: Garden Gear Online, Pinterest

Figures 7, 8. *Stepping stone paths; ideally stones would be different heights, as shown in Figure 8, creating an obstacle course activity.*

Photo Credit: DigsDigs

Figure 9. *Climbing wall; ideally would be on building wall and would include plants, as in Figure 10.*

Environmental Education

Embedding nature into MBCs also plays an important role in fostering environmental learning and appreciation in tweens. When people interact more with nature, they are more educated on the components of their local environments. This makes them care more about their environments (Garrard et al., 2017, p. 4).

The buildings surrounding the MBCs south of Broadway have plenty of potential to be retrofitted with natural elements. Because there are south-facing buildings in these two corridors, greenery can be added to building walls which get sufficient sunlight (*Figure 10*). While green wall maintenance can be problematic in dry months, rainwater can certainly water the plants in wet months. The combined incentives of stormwater storage, carbon sequestration, cooler air, and visual stimulation are significant (Beatley, 2011, p. 122). Plants like climbing vines could be good options to include here, and these would tie into the green climbing walls mentioned previously. Considering the relatively narrow width of the MBCs and the fact that these pathways are intended to carry much pedestrian traffic, it is unrealistic to expect large shrubs or trees to fit within them. Additionally, the quantity of sunlight that would reach trees in MBCs may be insufficient. Green walls are a creative yet subtle alternative to increasing nature in these corridors.

Photo Credit: gardeningknowhow.com

Figure 10. *Green wall.*

In a similar fashion, roofs of buildings surrounding MBCs can be fitted with greenery. In place of the typical grass- and soil-covered green roof (Beatley, 2011, p. 119), garden plots growing produce can be planted (*Figure 11*). These could provide fresh food for the local restaurants on Broadway (located across the street from MBCs), or for residents in nearby apartment buildings (Beatley, 2011 pp. 100-101). To attract tweens to these gardens, signage geared toward them (in the style of a scavenger hunt, perhaps) should be added into the MBCs, directing tweens to building entrances. The adventure to and unique destination of the gardens will be an interesting feature of the neighbourhood.

Once at the gardens, tweens can learn about plants and food systems in real life. Interactive game-like signage about gardening, growing one's own food, and food security will encourage further conversations on these topics. In addition to learning about the functioning of the gardens and related topics, tweens will get to witness how the plots attract local pollinators, like bees, butterflies, and birds, enhancing biodiversity. These gardens could offer workshops or planting days for tweens to reinforce their learning, too; these could be supported by the neighbourhood's library, adding on to its benefits, explored earlier by Gill. As tweens are the world's next generation of adults, the opportunity to become curious about environmental issues and protection is crucial, as they are inheriting a world with countless climate change-induced environmental issues. If they learn to care for their environments on a small scale, they may gather interest and skills required for tackling global crises in the

future.

Photo Credit: Instagram: @riverparknyc

Figure 11. *Rooftop garden with plenty of diverse plants.*

Adding green oases to sides and tops of buildings bordering MBCs is a way to increase tweens' time spent in nature and their involvement in local food systems. This also provides them with more places to relax and recharge their brains, benefitting both social well-being and mental wellness. These ideas are emphasized further in Lee's section.

Conclusion

Redeveloping MBCs to be nature-filled escapes from urban life is key to embedding connection with nature into teens' lives. Communities must have locations for tweens to gather inexpensively, so all the ideas proposed above should be free or low cost (see p. 20, subsection "Additional Park and Recreational Facility," in the chapter by Gill). A limitation of this report, however, is that it does not discuss funding for design ideas. The combination of the ideas presented in this section with those from other sections of this report gives South Granville Station's MBCs potential to be vibrant hubs for suburban tweens in Vancouver.

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PROMOTING ENVIRONMENTAL EQUITY

Geoffrey Lee

Promoting Environmental Equity in Mid-Block Connections Around South Granville Station

Executive Summary

Enhancing Mid-Block Connections (MBCs) and transit stations addresses environmental inequities while improving urban spaces for adolescents aged 12 to 14. By using green infrastructures, plants and sound-absorbing materials, to enhance air quality and lessen noise, it addresses the threats that air and noise pollution pose to mental health. These initiatives are complemented by sophisticated filtration systems, indoor green areas, and noise reduction in train stations producing serene and encouraging surroundings. These areas seamlessly integrated and constructed, set an example for inclusive and equitable urban planning by promoting social engagement, mental wellbeing, and environmental stewardship.

Introduction

Enhancing urban environments better suits the needs of teenagers aged 12 to 14, improving their social and mental wellbeing. Urban compared to rural settings are associated with a higher risk of serious mental health issues among adolescents (Buttazzoni, A., Doherty, S., & Minaker, L., 2021). Mid-block connections (MBCs) are essential for accomplishing this objective. They establish inclusive, accessible routes connecting communities to transit hubs. As an essential element of urban infrastructure, MBCs promote greener, healthier, and more engaging areas enhancing and promoting environmental fairness. This conversation broadens the scope to include the transport stations that MBCs connect to, building on the emphasis of incorporating peaceful aspects and natural components into MBCs. A cohesive strategy guarantees a smooth transition between MBCs and stations, providing teenagers with a nurturing atmosphere. Addressing the negative effects of environmental injustices like pollution on mental health, incorporating green areas into stations and passageways, and developing comprehensive plans to build fair, supportive urban experiences are all important factors in achieving this goal.

The Impacts of Environmental Inequities on Adolescent Mental Health

Adolescents are a particularly susceptible group due to their developmental period. Environmental injustices such as exposure to air and noise pollution have a significant impact on their mental health. Long-term exposure to nitrogen dioxide (NO_2) and fine particulate matter (PM2.5) has been shown to dramatically raise the risk of anxiety, depression, and cognitive decline (Lyons, S., Nolan, A., Carthy, P., Griffin, M., & O'Connell, B., 2024). Cleaner urban settings are urgently needed, as evidenced by a study published in *JAMA Network Open* showing a robust correlation between early exposure to air pollution and the emergence of mood and psychotic illnesses in young people. These problems are made worse by noise pollution, as prolonged exposure to transit and traffic noise impairs sleep, raises stress hormones, and exacerbates depressive and anxious symptoms. Research conducted in urban areas such as London demonstrated that combined effects of noise and air pollution have a compounding effect on the emotional and cognitive development of adolescents (Thompson et al., 2024). These results indicate the twofold load of environmental stresses that young people in crowded, transit-heavy neighbourhoods like South Granville must contend with.

Environmental inequities must be addressed to ensure equal access to clean air, green spaces, and restorative environments. Mid-block connections (MBCs) and transit stations play pivotal roles in mitigating environmental stressors. Incorporating green walls, trees, and vegetation into MBCs will effectively filter pollutants, improving air quality. Green walls have been shown to absorb particulate matter and other pollutants, enhancing urban air quality (Vitaliano, S., Cascone, S., & D'Urso, P. R., 2024), moreover, trees and shrubs contribute to the removal of airborne pollutants (Currie, B. A., & Bass, B. ,2008). By installing these filtration systems, transit stations can significantly improve the quality of the air. Stations can even greatly lower noise levels by utilising sound-absorbing materials, which will make passengers' surroundings more tranquil. To enhance interior acoustic comfort in railway carriages, new materials and designs have been created; these can also be used in station settings (Global Railway Review, 2021). Such strategies create supportive, caring environments that put adolescent mental health first, allowing young people to move through cities stress-free.

Recommendations for a Journey-Wide Approach to Environmental Equity

Achieving environmental fairness in transit trips requires a coordinated strategy that incorporates green spaces, peaceful elements, and educational programs in transit stations and mid-block connections (MBCs). Since adolescents are extremely vulnerable to environmental stressors like noise and air pollution, it is crucial to design transit spaces that both address these issues and support mental health. Urban design may improve the entire transportation experience and reduce environmental disparities by converting MBCs and stations into connected, supporting areas. According to *Medical News*, early childhood exposure to air and noise pollution is linked to a higher risk of anxiety and other mental health problems in adolescence and early adulthood.

Enhancing MBCs for Environmental Equity

MBCs serve as critical corridors linking neighborhoods to transit stations, and their design has a significant impact on users' mental well-being and comfort. Incorporating green infrastructure into MBCs is one of the most effective strategies for addressing air pollution while creating a visually appealing and restorative environment. Plants like peace lilies and spider plants are well-known for their air-purifying capabilities, effectively removing nitrogen dioxide (NO_2) and particulate matter (PM2.5) from the air. Vertical green walls, adorned with climbing vines, can be strategically placed along MBCs to act as natural air filters while adding texture and life to the built environment. These green elements not only improve air quality but also provide a calming visual experience, supporting mental wellness through exposure to nature's restorative effects (Newbury et al., 2024).

Source: https://www.treehugger.com/

Figure 1: *Trees acting as noise barriers*

Source:www.grammbarriers.com

Figure 2: *Green barriers*

In addition to improving air quality, reducing noise pollution within MBCs is crucial. Urban noise from traffic and transit systems exacerbates stress, disrupts sleep, and impairs cognitive function, particularly in adolescents. To address this, sound-absorbing materials such as recycled rubber pathways or noise-dampening panels can be incorporated into MBC design such as the examples in *Figure 1 and 2*. These features create quieter, more relaxing spaces for pedestrians and cyclists. Green barriers, such as dense shrubs or trees, can further mitigate noise while reinforcing the corridor's natural aesthetic. Studies show that integrating such noise-buffering interventions reduces stress levels, making these corridors more inviting and comfortable (Thompson et al., 2024).

Mid-block connections (MBCs) to transit stations should be smoothly connected by wayfinding systems to provide a seamless passenger experience. Signage centred on local wildlife or vegetation can be used to reinforce the journey's environmental purpose while providing passengers with straightforward guidance. By incorporating QR codes into these signs, teens are encouraged to recognise and engage with the green aspects of the MBCs through interesting and instructive material. This strategy encourages environmental awareness in addition to improving user engagement (Qrlab, n.d.). A seamless transition is promoted by keeping the design aesthetic of MBCs and station entrances similar, enabling users to travel with confidence and ease. In transit systems, efficient wayfinding has been demonstrated to enhance both overall ridership and user experience (The science of wayfinding—a formula to boost transit ridership, n.d.).

Integrating Calming Features in Stations

Transit stations play an equally vital role in addressing environmental inequities and enhancing user well-being. Improving air quality inside stations is paramount, particularly in dense urban areas where pollutants accumulate. Cutting-edge air filtration systems, such as high-efficiency particulate air (HEPA) filters or activated carbon filters, can remove harmful particles like PM2.5 and NO_2, ensuring a healthier environment for station users. This is especially important for adolescents, whose developing lungs and brains are more vulnerable to the negative effects of air pollution (Newbury et al., 2024).

Stations can include indoor green areas that mimic the vegetation in nearby MBCs to go along with these air quality enhancements. Small indoor parks, plant-filled atriums, or rooftop gardens offer spaces for socialising and rest, especially for teenagers juggling the

demands of city life. By linking these areas to the MBCs' greenery, continuity is guaranteed and a cohesive sensory experience is produced. Specifically, rooftop gardens can be used as multipurpose areas that support local biodiversity and host community events like gardening classes or environmental education seminars.

Noise mitigation is another critical consideration for station design. High-noise areas such as platforms and ticketing zones can be outfitted with soundproofing materials like acoustic panels or barriers. These features significantly reduce noise levels, creating calmer, less overwhelming environments for young transit users. By minimizing auditory distractions, stations can foster a more relaxing atmosphere, encouraging positive social interactions and reducing stress.

Source: https://www.precious-learners.com/

Figure 3. *Nature inspired youth classes*

Social and Educational Initiatives

Promoting a feeling of pride and ownership in transportation areas requires active community involvement. Adolescents can be empowered to engage with their environment through youth-focused activities such as gardening workshops, art classes inspired by nature, *Figure 3*, or environmental stewardship programs held in both MBCs and station green areas. Collaborating with nearby libraries or schools to develop activities like rooftop garden planting days offers practical chances for education and teamwork. The transit experience is further enhanced by interactive educational displays. Information regarding mental wellbeing, environmental equity, and the green aspects of the transport areas can be shared via digital kiosks or signage that can be scanned by QR codes (Xiao, M., 2024).

These displays offer an engaging way to learn about sustainable practices and the role of green infrastructure in urban design. Education initiatives not only deepen users' appreciation for their surroundings but also inspire future environmental stewardship.

Creating a Unified Journey

For environmental fairness to be implemented journey-wide, MBC and station designs must be in harmony. Calm elements such as lavender-lined walkways in MBCs can create a more tranquil travel experience, emphasis by my colleague Natasha Ivkov.. By incorporating these ideas into stations with rooftop gardens or interiors brimming with plants, consumers are guaranteed a seamless experience from beginning to end. This coordinated strategy enhances the advantages of green infrastructure for mental and social well-being and fosters a supportive atmosphere for teenagers as they progress.

Urban planners can create fair, supportive environments that address environmental injustices and promote mental health by including these suggestions into MBCs and stations. In addition to helping teenagers, this strategy establishes a benchmark for sustainable, inclusive urban planning.

Conclusion

A journey-wide approach to environmental equity emphasizes the interconnected roles of mid-block connections (MBCs) and transit stations in fostering adolescent mental wellness. MBCs can mitigate air and noise pollution through green infrastructure like climbing vines and sound-dampening materials, while cohesive wayfinding systems enhance accessibility and continuity. Transit stations complement these efforts with cutting-edge air filtration systems, indoor green spaces, and soundproofing, creating an environment where adolescents can feel supported and thrive. By skilfully incorporating relaxing elements throughout the transit trip, these initiatives collectively alleviate environmental injustices and foster mental wellness. These integrated solutions must be given top priority by urban planners and legislators in order to address the negative impacts of pollution and urban pressures on young people. The renovation of South Granville Station presents a singular chance to establish a standard for inclusive and sustainable transit architecture. This project can act as a model for equitable urban development by putting green infrastructure into place and encouraging community involvement through programs like gardening workshops and educational displays.

The strategy presented here may be used as a model for upcoming urban planning and transportation initiatives. In addition to being a question of justice, ensuring fair access to restorative settings is also an investment in the general public's health and wellbeing. Cities can build transit systems that are not only practical but also genuinely supportive of all demographics by tackling environmental injustices and placing a high priority on mental wellness. This will promote healthier, more cohesive communities for future generations.

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Concluding Remarks

The MBCs around South Granville Station are unique corridors with the potential to serve as gathering places for pre-teens and young teens, reducing the barriers they face to taking public transportation. Among those barriers is a lack of design supporting mental wellness and social well-being in areas around public transport, in the subcategories discussed throughout. Design ideas proposed in each section of this report serve as starting points for redevelopment of South Granville Station's MBCs for the user group, ultimately increasing sustainability through public transport use. These ideas can be applied to further urban design projects in Metro Vancouver, which must take the needs of all demographic groups (including pre-teens and young teens) into consideration.